How Wine Is Made

From vine to bottle — the winemaker's road for each style of wine. Tap any step to open the details, hover the **underlined terms** for a quick definition.

Red Wine

1. Harvesting *Vendange*

Grapes are picked either **by hand** — gentler, and the only option on steep or terraced slopes — or **by machine**, which is faster and cheaper. Hand-picking allows a first selection of healthy bunches right in the vineyard.

Why it matters: picking date and method set the sugar, acidity and health of the fruit — everything downstream depends on it.

2. Sorting *Triage*

On a **sorting table** (or optical sorter) unripe, raisined or rotten berries and leaves and other debris (MOG) are removed so only sound fruit continues.

Why it matters: a single handful of rot or green fruit can taint a whole vat.

3. Crushing & Destemming OPTIONAL *Foulage & Éraflage*

The **crusher/destemmer** splits the berries to release juice and, usually, strips them off the stalks. Some reds are fermented as **whole bunches** instead (carbonic maceration in Beaujolais, whole-cluster in the Rhône and Burgundy). A little **sulphur** may be added here.

Why it matters: stems add grip and a peppery lift when ripe, but green tannin when not — hence ‘optional’.

4. Maceration + Fermentation *Cuvaison*

This is the heart of red winemaking. In the vat, yeast turns sugar into alcohol **while** the juice steeps on the skins — the stage called **cuvaison**. The floating cap of skins is worked back in by **pigeage** (punching down) or **remontage** (pumping over) to extract colour and tannin. Temperature is controlled; it can run 5–21 days.

Why it matters: almost all of a red wine's colour, tannin and structure is built here.

5. Pressing OPTIONAL *Pressurage*

The **vin de goutte** (free-run) drains off first; the leftover **marc** is then pressed to extract the darker, firmer **vin de presse**. The winemaker decides how much press wine, if any, to blend back.

Why it matters: press wine adds colour and grip — a little can add depth, too much brings harshness.

6. Malolactic Fermentation *Fermentation malolactique*

A second, **bacterial** fermentation converts sharp malic acid into softer lactic acid. For reds this is essentially always done, and it also stabilises the wine.

Why it matters: it rounds a tart young wine and prevents the change happening later, in bottle.

7. Barrel Ageing OPTIONAL *Élevage*

A **vin de garde** is matured (**élevage**) in oak **barriques** or larger **foudres**, sometimes for up to two years. New oak adds spice and tannin; older wood mainly lets the wine breathe and settle.

Why it matters: oak and time knit the tannins together and add complexity — but everyday wines skip it to stay fresh and affordable.

8. Fining & Filtering OPTIONAL *Collage & Filtration*

To polish the wine before bottling it may be **fined** — classically with **egg white** (**collage**) — and/or **filtered**. Both remove haze and harsh particles. Some producers do neither, bottling 'unfined and unfiltered'.

Why it matters: it makes the wine bright and stable, though heavy filtering can strip a little texture.

9. Bottling *Mise en bouteille*

The finished wine is bottled, corked or capped and labelled. Vins de garde may then rest months or years more before release.

Why it matters: the journey from vine to glass is complete — the rest is patience.

White Wine

1. Harvesting *Vendange*

As for red — hand or machine — but whites are frequently picked **earlier** and in the **cool of the morning/night** to preserve fresh acidity and aroma.

Why it matters: a white's zip and perfume are easy to lose to over-ripeness or heat.

2. Sorting *Triage*

Damaged and unripe berries and debris are removed before pressing.

Why it matters: with no skin-contact to hide behind, faults show quickly in white wine.

3. Crushing / Destemming OPTIONAL *Foulage / Éraflage*

Whites are often **whole-bunch pressed** with no crushing at all, for the cleanest juice. When crushing is used it is gentle, and skin contact is deliberately brief or skipped.

Why it matters: long skin contact would add colour and phenolic bitterness that most whites don't want.

4. Pressing *Pressurage*

Unlike red, whites are **pressed before fermentation** — the skins are separated straight away and only the juice goes forward. Any leftover **residue** can be used for basic table wine.

Why it matters: pressing before fermentation is *the* defining difference between white and red winemaking.

5. Débourbage / Clarification *Débourbage*

The cloudy juice is chilled and left so the coarse solids ('bourbes') drop out; the clean juice is racked off the sediment before fermenting. A little **sulphur** protects it meanwhile.

Why it matters: fermenting clean juice gives a purer, more precise wine.

6. Fermentation *Fermentation alcoolique*

Yeast converts sugar to alcohol, usually at **cool temperatures** (12–18 °C) in tank — or, for the Grands Vins (white Burgundy, top Bordeaux), in oak barrel.

Why it matters: cool, slow fermentation preserves the fragile floral and fruity aromas.

7. Malolactic Fermentation OPTIONAL *Fermentation malolactique*

Not automatic for white wine. It is **encouraged** for richer, oaked styles (white Burgundy) to add a creamy roundness — and **blocked** for crisp, aromatic styles (Sauvignon, Riesling) to keep them zesty.

Why it matters: the single biggest stylistic choice between a 'buttery' and a 'crisp' white.

8. Fining & Clarification OPTIONAL *Collage*

Whites are usually fined — commonly with **bentonite** clay to remove unstable proteins — and clarified/filtered so they stay bright and won't throw a haze.

Why it matters: a white showing haze on the shelf looks faulty, even if it tastes fine.

9. Bottling *Mise en bouteille*

Most whites are bottled and released young to capture their freshness; a few age-worthy whites are held back.

Why it matters: freshness is the currency of most white wine — speed to bottle protects it.

Sweet & Late-Harvest

1. Concentrating the Grapes *Surmaturité*

Sweetness starts in the vineyard by super-concentrating the sugar: **Botrytis cinerea** (Noble Rot) shrivels the berries (Sauternes, Tokaji, Alsace SGN); **passerillage** dries them on the vine or on straw; or the grapes are left to **freeze** for ice wine.

Why it matters: there is no way to fake this — great sweet wine is made from a fraction of the normal crop.

2. Selective Harvest *Tries successives*

Botrytised grapes ripen unevenly, so pickers go through the vineyard **several times** (*tries*), taking only the shrivelled berries at their peak. Painstaking and tiny in yield.

Why it matters: this hand labour is why the greatest sweet wines are so rare and costly.

3. Pressing *Pressurage*

The dehydrated grapes give up very little juice, and slowly. What emerges is intensely sugary must.

Why it matters: yields can be a quarter of a dry wine's — concentration comes at the cost of quantity.

4. Fermentation (partial) *Fermentation partielle*

Yeast begins converting sugar to alcohol but **cannot finish** — the sheer sugar and rising alcohol exhaust it. Fermentation is deliberately **stopped** by chilling and adding **sulphur** while plenty of natural sweetness remains.

Why it matters: the balance of alcohol, sugar and acidity here defines the whole wine.

5. Stopping & Stabilising OPTIONAL *Mutage / sulfitage*

The remaining yeast is removed by chilling, centrifugation, pasteurisation or racking so it can't restart. (For fortified *vins doux naturels*, fermentation is instead halted by adding grape spirit — **mutage**.) Residual sugar typically lands around **50–150 g/L**.

Why it matters: a bottle that re-ferments would be spoiled — stability is essential for sweet wine.

6. Ageing & Bottling *Élevage & mise*

Many sweet wines are **élevage**-aged (often in barrel) and are famously long-lived, evolving for decades in bottle.

Why it matters: high sugar and acidity act as natural preservatives — these are among the most age-worthy wines of all.

Rosé

1. Harvesting & Sorting *Vendange & Triage*

Rosé is made from **black (red) grapes**, usually picked to keep bright acidity, then sorted as for any quality wine.

Why it matters: rosé lives on freshness, so healthy, crisp fruit is key.

2. Brief Skin Contact *Macération courte*

Colour comes from a **short** maceration on the skins — just a few hours. Two routes: **direct press** (pressed almost like a white, very pale — the Provence style) or **saignée**, where pink juice is 'bled' off a red-wine vat.

Why it matters: the length of skin contact is exactly what sets a rosé's colour, from palest pink to vivid.

3. Pressing / Running-off *Pressurage / Saignée*

The lightly-coloured juice is separated from the skins. In the **saignée** method the red wine left behind is concentrated as a by-product.

Why it matters: note EU rules forbid blending red + white to make still rosé — the colour must come from the skins.

4. Cool Fermentation *Fermentation*

From here rosé follows the **white-wine road**: the juice ferments cool in tank to keep its delicate red-fruit and floral aromas.

Why it matters: temperature control protects the fresh, fruity character rosé is loved for.

5. Fining & Bottling OPTIONAL *Collage & Mise*

Lightly fined and clarified, then bottled early. The vast majority of rosé is made to be drunk within a year, chilled.

Why it matters: rosé is at its best young and fresh — most isn't meant to age.

In the winemaker's words

Making Red Wine

After the harvest, the grapes are separated from the stalks and sorted — manually or mechanically — to remove unwanted fruit and other material. The fruit is then crushed and usually destemmed. At that point a little **sulphur** may be added to prevent early oxidation of the **must**, which is then pumped into stainless-steel tanks or wooden vats (**foudres**).

For red wine, skin contact and alcoholic fermentation happen **together** in the same vat — a stage the French call **cuvaison**. Over several days to a few weeks the cap of skins is kept in contact with the juice, by **pigeage** (punching down) or **remontage** (pumping over), to draw out colour and tannin. When fermentation finishes, the free-run **vin de goutte** is run off and the leftover **marc** is pressed to yield **vin de presse**, which may or may not be blended back.

The wine then rests while **malolactic fermentation** softens its acidity. A **vin de garde** may be barrel-aged (**élevage**) for up to about 24 months before it is fined — often with egg white (**collage**) — and/or filtered, then bottled.

Making White Wine

The huge variety of white wines reflects many different vinifications, each adapted to a particular wine. White winemaking differs from red at several stages — above all, the skins are rarely **macerated** with the juice. Instead the grapes are usually pressed straight away and only the juice is fermented.

After pressing, the cloudy juice is left to settle so the solids drop out — a clarification step called **débourbage**. Fermentation then runs cooler than for red, to protect delicate aromas. Most whites are not barrel-aged, though some Grands Vins — the grands crus of Burgundy, the classified wines of Bordeaux — are raised in oak. **Malolactic fermentation** is not automatic for white wine but is increasingly used for the Grands Vins. A final fining (often with **bentonite**) and clarification precede bottling.

Making Sweet & Late-Harvest Wine

Sweet wines keep sugar in the finished wine by starting with far riper, more concentrated grapes and then stopping fermentation before the yeast has eaten all the sugar. Concentration can come from **Botrytis cinerea** (Noble Rot), which shrivels the berries — as in Sauternes, the Loire's Quarts-de-Chaume, Alsace Sélection de Grains Nobles, or Tokaji — from drying on the vine (**passerillage**), or from freezing (ice wine / Eiswein).

Fermentation is halted by chilling, by adding **sulphur** dioxide, and by removing the yeast through centrifugation, pasteurisation or racking. The resulting wines are extremely rich — roughly 50–150 g/L of residual sugar. (A related family, the *vins doux naturels* such as Muscat de Beaumes-de-Venise, instead stop fermentation by adding grape spirit — a step called **mutage**.)

Making Rosé

Rosé takes its colour from brief skin contact, not from blending red and white (forbidden for still wine in the EU, with Champagne the notable exception). Two main methods: **direct press**, where red grapes are pressed almost like whites for a very pale colour (the Provence style); and **saignée** ("bleeding"), where a little pink juice is run off a red-wine vat after a few hours of maceration, giving a deeper colour — and incidentally concentrating the red wine left behind. Fermentation and the rest follow the white-wine road.

Glossary

Bentonite
A natural clay used to fine white wines: it grabs unstable proteins that would otherwise cloud the wine, then settles out.

Botrytis cinerea — Noble Rot
A benign fungus that, in the right damp-then-dry autumn, shrivels grapes and concentrates their sugar and flavour — the secret of Sauternes, Tokaji and the great sweet wines.

Collage — fining
Adding a fining agent (egg white, bentonite clay, gelatine…) that binds to haze and harsh particles and drags them to the bottom, clarifying and softening the wine.

Cuvaison
The red-wine period when juice, skins and pips macerate together **while alcoholic fermentation is happening** — this is where colour and tannin are extracted.

Débourbage
Settling freshly pressed white juice so the coarse solids ('bourbes') drop out and clean juice can be racked off before fermentation.

Élevage
Literally 'raising' — the ageing/maturation of the wine after fermentation, in barrel, foudre or tank, until it is ready to bottle.

Foudre
A large wooden vat (much bigger than a barrel), used for fermenting or ageing without adding much oak flavour.

Grand Vin
A producer's top wine — given the ripest fruit and the most careful (and slowest) élevage.

Maceration
Steeping the juice on the grape skins to extract colour, tannin and flavour. Long for reds, brief or skipped for whites, short for rosés.

Malolactic fermentation — *FML*
A second, bacterial fermentation that converts sharp malic acid into softer lactic acid. Standard for reds; optional for whites, where it adds a rounder, 'buttery' note.

Marc
The solid cake of skins, pips and pulp left after pressing (the 'pomace'). Can be distilled into the spirit also called *marc*.

Must — *moût*
Freshly crushed or pressed grape juice, with (for reds) the skins, pulp and pips, before and during fermentation.

Mutage
Stopping fermentation by adding neutral grape spirit; it kills the yeast while sugar remains, making fortified *vins doux naturels* and Port-style wines.

Passerillage
Drying the grapes to concentrate sugar — left on the vine past ripeness, or on straw mats — without Noble Rot. The basis of many *vins de paille*.

Pigeage
'Punching down' — pushing the floating cap of skins back into the juice by hand or paddle to keep colour and tannin extracting.

Remontage
'Pumping over' — drawing juice from the bottom of the vat and spraying it over the cap of skins; the mechanical equivalent of pigeage.

Sulphur (SO_2)

Sulphur dioxide, the winemaker's main antioxidant and antiseptic. A little protects juice and wine from oxidation and stray microbes; ‘natural’ wines aim to use as little as possible.

Vin de garde

A wine built to age — with the structure (tannin, acidity, concentration) to improve over years in bottle, as opposed to an early-drinking wine.

Vin de goutte

The ‘free-run’ wine that flows out of the vat under its own weight, before pressing. Usually the finest fraction.

Vin de presse

The wine squeezed from the leftover skins (the marc) after the free-run has drained. Darker and more tannic; blended back in part, or kept separate.

Interactive edition of *“How Wine Is Made”* · A Quick Glance at French Wine · Chris Paul.
Works fully offline — a single self-contained file.